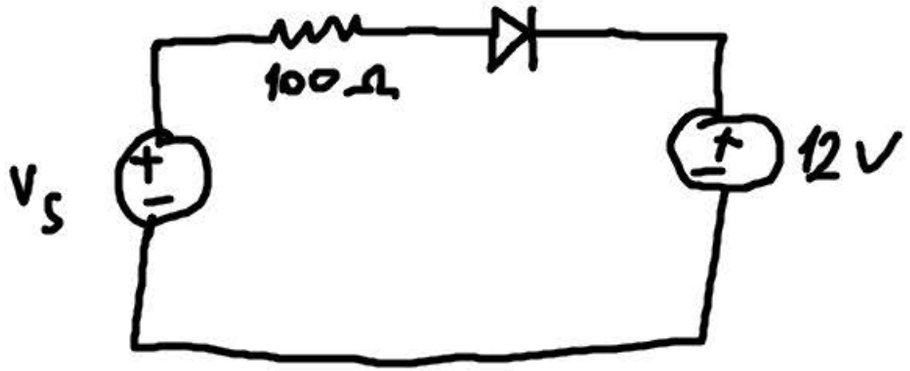
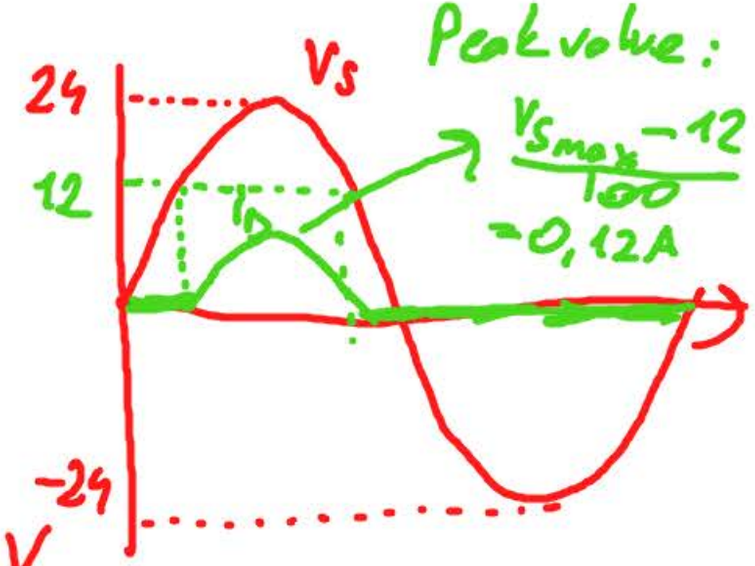
Ideal Diode Exercises

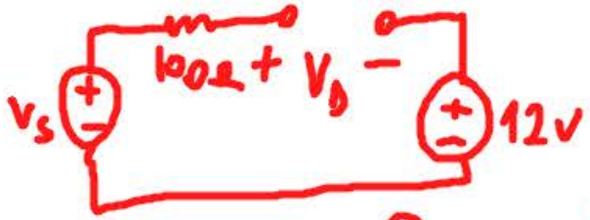
1) Consider the following circuit consisting of an ideal diode.
Given $v_S = 24\sin\omega t$, sketch the waveform of i_D.





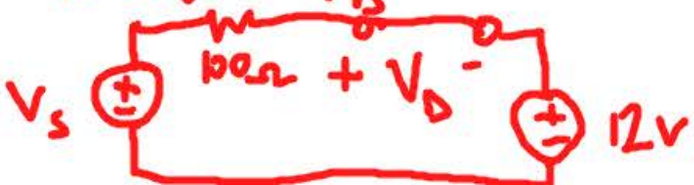
Solution

Diode is off when $V_D < 0 \Rightarrow$ when $V_S < 12V$



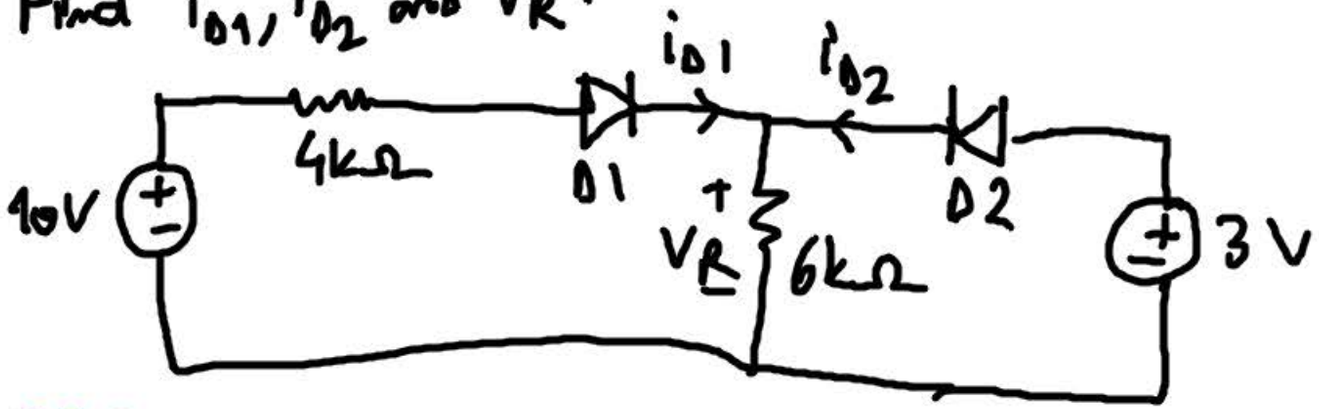
In this case $i_D = 0$ A

Diode is ON when $V_D = 0 \Rightarrow$ when $V_S > 12V$



In this case $i_D = \frac{V_S - 12}{100}$ A

2) Consider the following circuit consisting of two ideal diodes.
Find i_{D1}, i_{D2} and V_R.



Solution:

In order to solve electrical circuits consisting of diodes, we make assumptions about the states of the diodes. If the solution is feasible, our assumption is correct and we have found the solution. If it is infeasible, we make a different assumption until we get a feasible solution

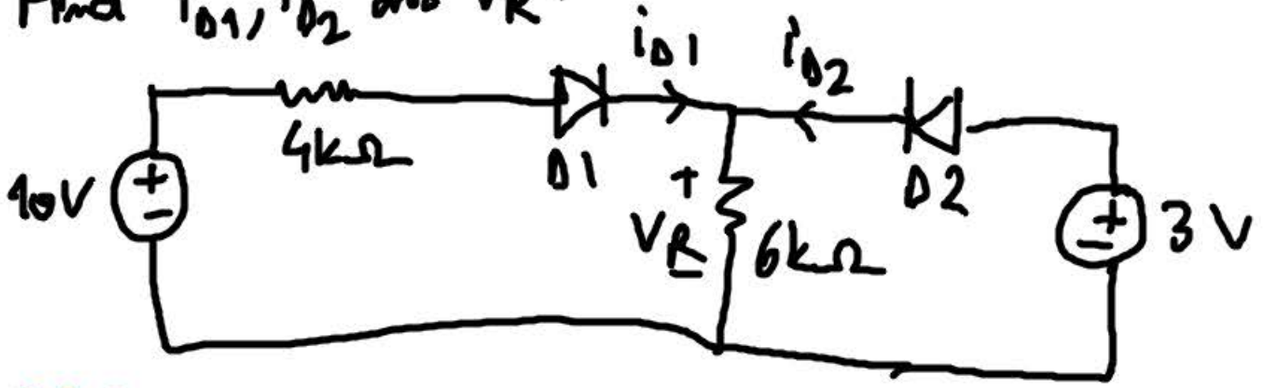
D1	D2
OFF	OFF
OFF	ON
ON	OFF
ON	ON

A) Assume D1 is OFF and D2 is OFF



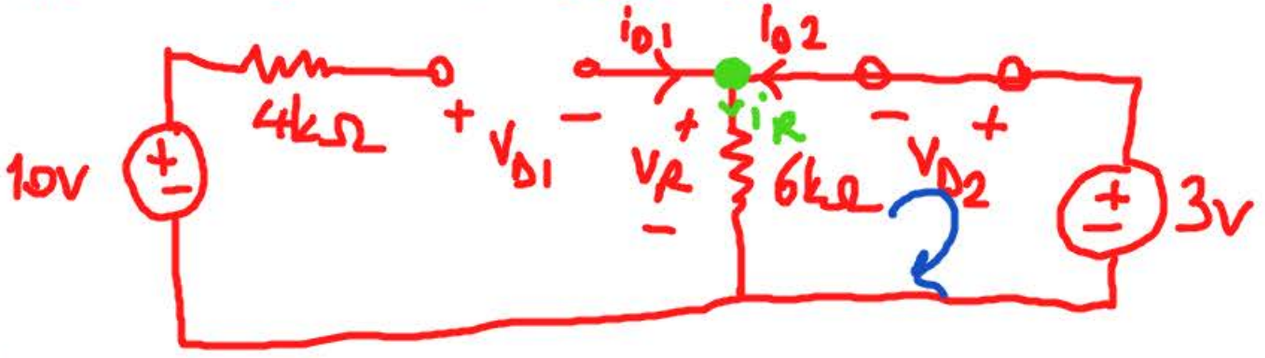
$V_{D1} = 10V$
$V_{D2} = 3V$
} Violates nonconducting diode state properties

2) Consider the following circuit consisting of two ideal diodes.
Find i_{D1}, i_{D2} and V_R.



Solution:

B) Assume D1 is OFF and D2 is ON.



D1	D
OFF	OFF
OFF	ON
ON	OFF
ON	ON

KCL: $i_{D1} + i_{D2} = i_R$

$i_{D1} = 0$

$i_{D2} = i_R$

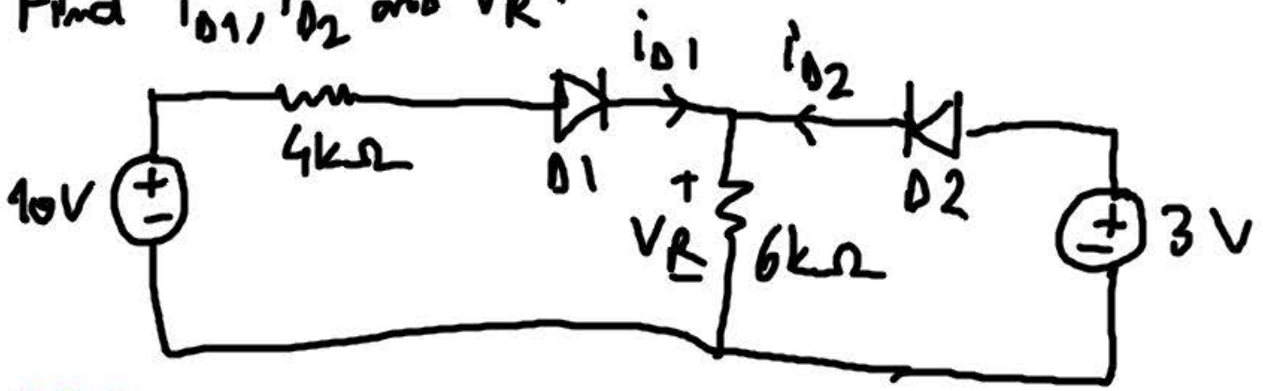
KVL: $-V_R + 3V = 0$

$V_R = 3V$

$V_{D1} = 10 - V_R = 7V > 0$

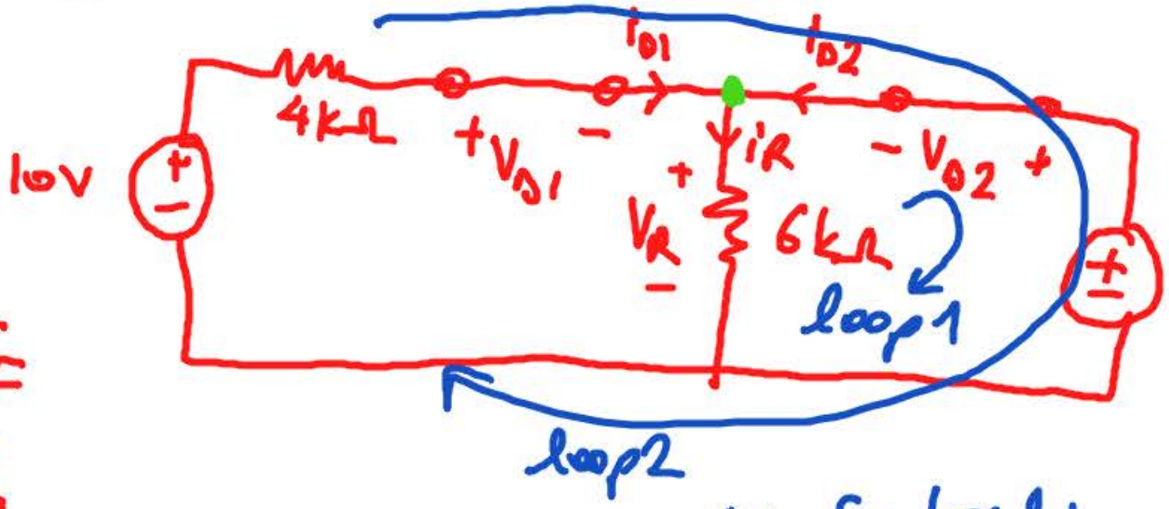
Violates the conducting state conditions of D1.

2) Consider the following circuit consisting of two ideal diodes.
Find i_{D1}, i_{D2} and V_R.



Solution:

C) Assume D1 is ON and D2 is ON.



D1	D
OFF	OFF
OFF	ON
ON	OFF
ON	ON

KCL: $i_{D1}+i_{D2}=i_R$

KVL for loop 2:

$-10+V_{R4}+3=0$

$V_{R4}=7V$

Ohm's law: $V_{R4}=i_{D1}\cdot 4k\Omega$

$i_{D1}=7/4k\ A$

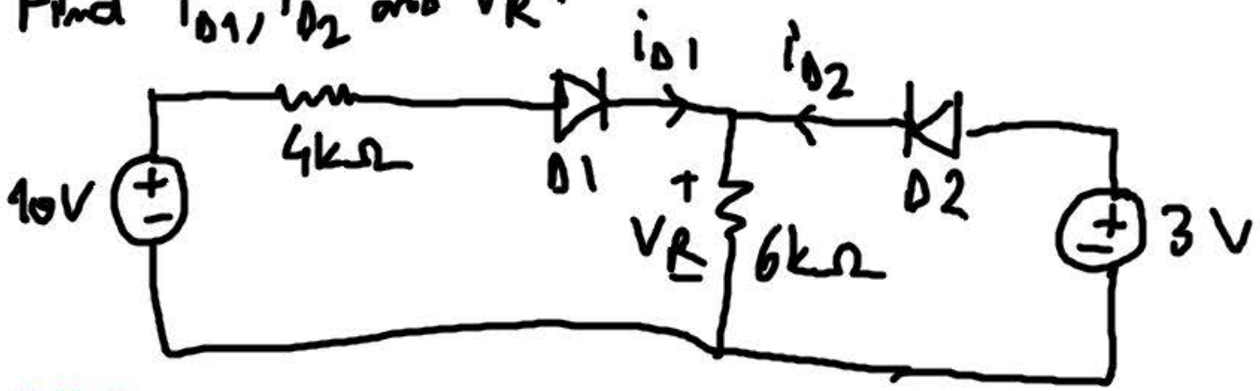
$i_{D2}=i_R-i_{D1}<0$ violates the cond

KVL for loop 1:

$-V_R+3=0 \Rightarrow V_R=3V$

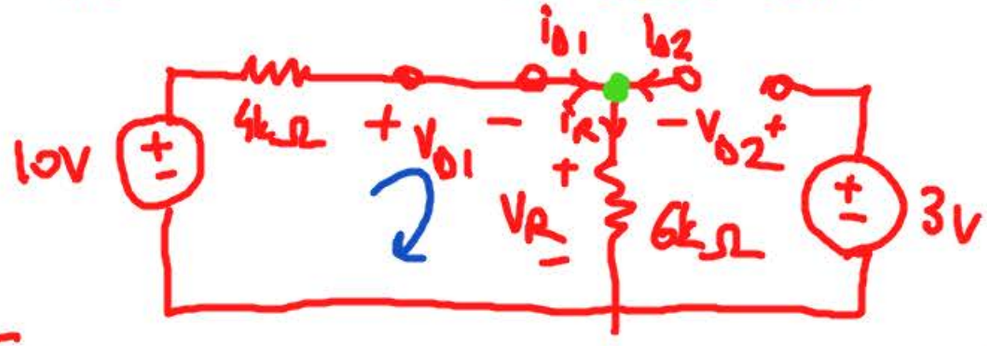
Ohm's law: $V_R=i_R\cdot 6k \Rightarrow i_R=\frac{3}{6k}A$

2) Consider the following circuit consisting of two ideal diodes.
Find i_{D1}, i_{D2} and V_R.



Solution:

D) Assume D1 is ON and D2 is OFF



D1	D
OFF	OFF
OFF	ON
ON	OFF
ON	ON

KCL: $i_{D1} + i_{D2} = i_R$

$i_{D2} = 0 \Rightarrow i_R = i_{D1}$

KVL: $-10V + V_{R4} + V_R = 0$

Ohm's law: $V_{R4} = i_{D1} \cdot 4k$

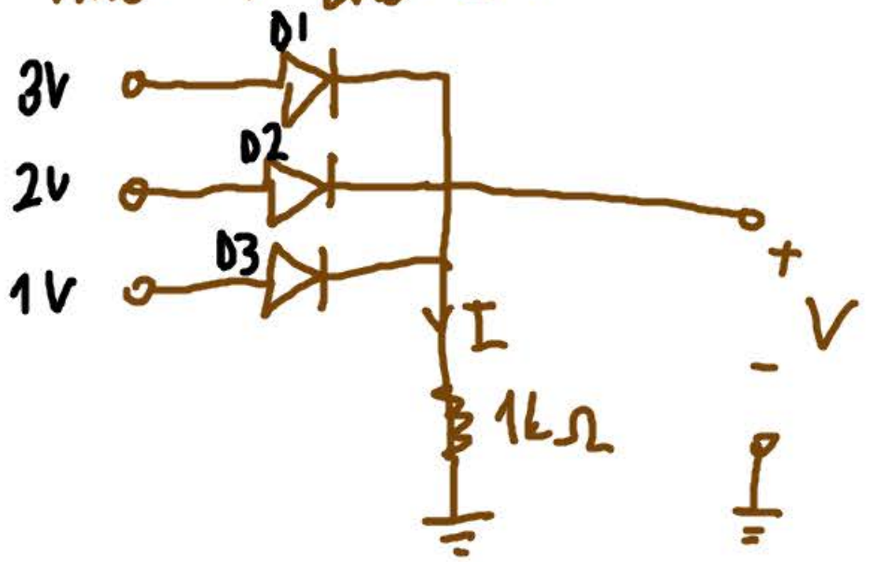
$V_R = i_R \cdot 6k = 6V$

$10 = i_{D1} \cdot 10k$

$i_{D1} = 1mA > 0$ ✓

$V_{D1} = 0$ ✓ $\quad V_{D2} = 3 - V_R = 3 - 6 \cdot 1 = -3V < 0$ ✓

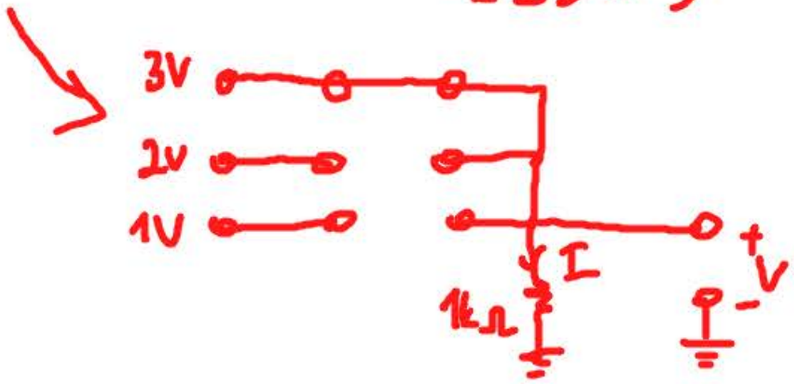
3) Consider the following circuit consisting of three ideal diodes.
Find V and I.



Solution:

D1	D2	D3
OFF	OFF	OFF
OFF	OFF	ON
OFF	ON	OFF
OFF	ON	ON
ON	OFF	OFF
ON	OFF	ON
ON	ON	OFF
ON	ON	ON

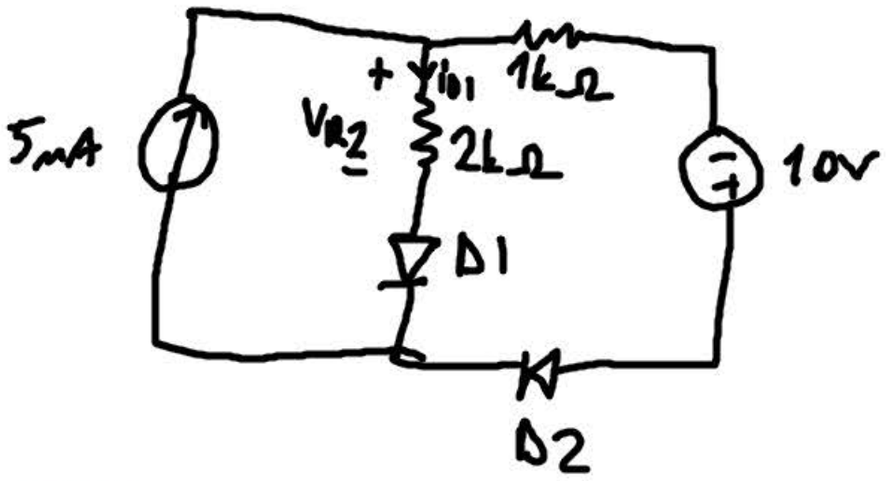
If D1 is ON, then D2 and D2 must be OFF
If D2 is ON, $V_{D1} > 0$ is infeasible ✗
If D3 is ON, $V_{D1} > 0$, $V_{D2} \geq 0$ } infeasible ✗



V = 3 Volts
$I = \frac{3}{1k} = 3\,mA$

4) Consider the following circuit consisting of two ideal diodes

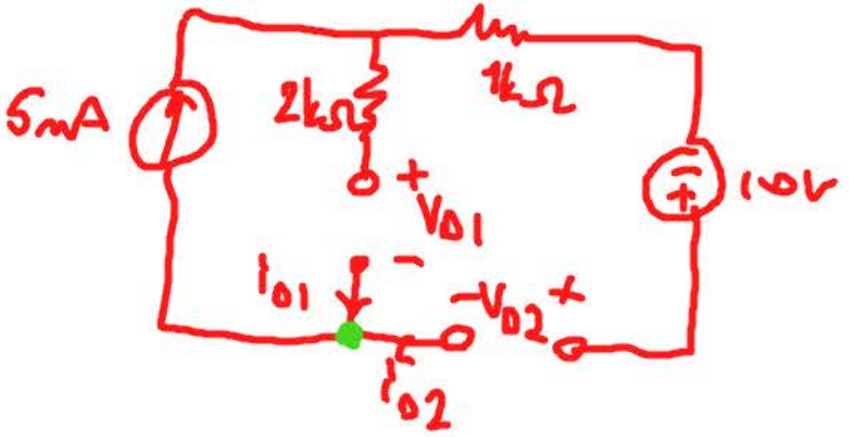
Find i_{D1} and V_{R2}.



Solution:

D1	D2
OFF	OFF
OFF	ON
ON	OFF
ON	ON

A) Assume D1 & D2 are OFF



$i_{D1} = 0$

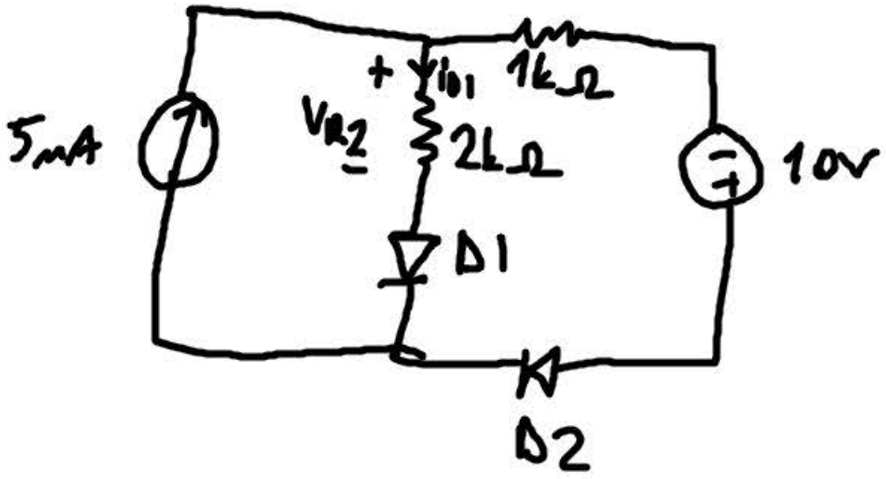
$i_{D2} = 0$

KCL: $i_{D1} + i_{D2} = 5mA$

$0 = 5mA$ ✗

4) Consider the following circuit consisting of two ideal diodes

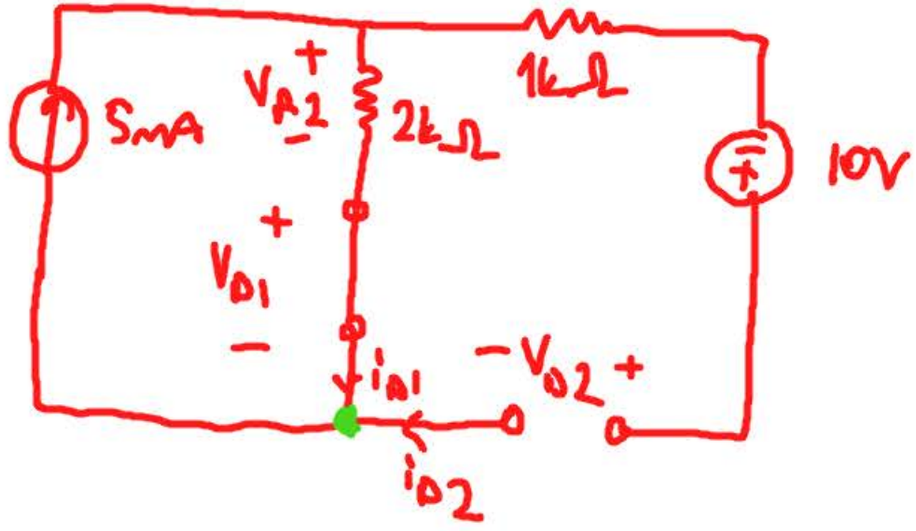
Find i_{D1} and V_{R2}.



Solution:

D1	D2
OFF	OFF
OFF	ON
ON	OFF
ON	ON

B) Assume D1 is ON and D2 is OFF



KCL: $5mA = i_{D1} + i_{D2}$

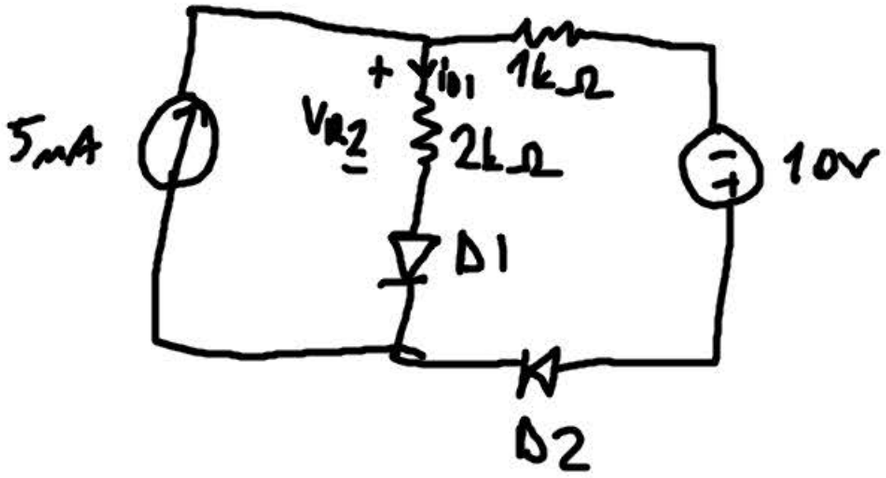
Since $i_{D2} = 0 \Rightarrow i_{D1} = 5mA > 0$ ✓

Ohm's law: $V_{R2} = i_{D1} \cdot 2k = 10V$

$V_{D2} = V_{R2} + 10V = 20V > 0$ ✗

4) Consider the following circuit consisting of two ideal diodes

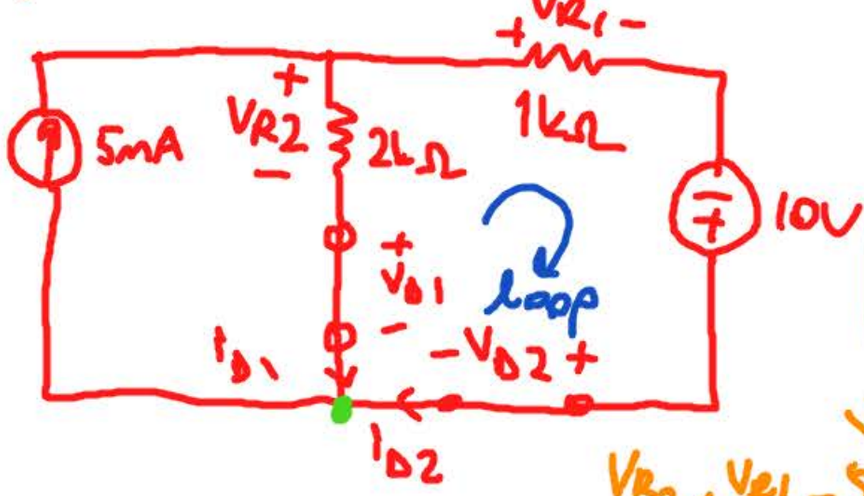
Find i_{D1} and V_{R2}.



Solution:

D1	D2
OFF	OFF
OFF	ON
ON	OFF
ON	ON

C) Assume D1 & D2 are ON



KCL: $i_{D1} + i_{D2} = 5mA$

KVL: $-V_{R2} + V_{R1} - 10 = 0$

Ohm's law: $V_{R1} = 1k \cdot i_{D2}$, $V_{R2} = 2k \, i_{D1}$ ⟹ $i_{D1} = V_{R2}/2k$, $i_{D2} = V_{R1}/1k$

$\frac{V_{R2}}{2k} + \frac{V_{R1}}{1k} = 5mA$

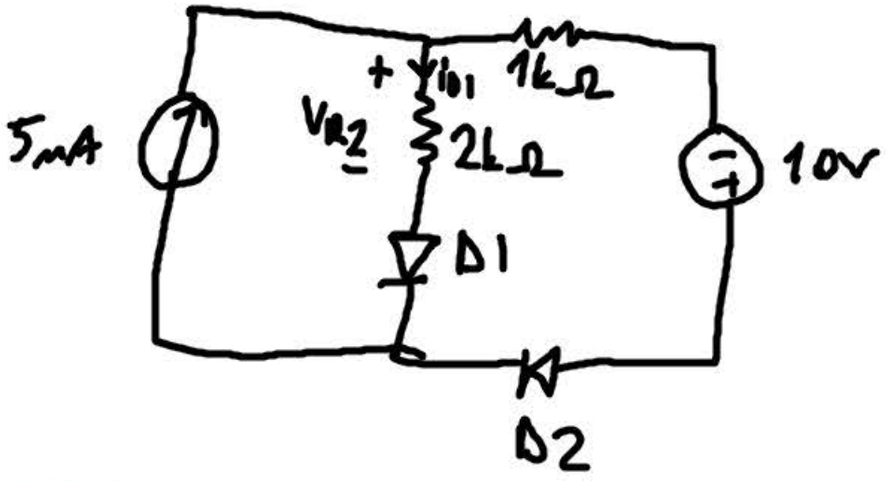
$V_{R1} + 2V_{R2} = 10V$ ⟶

$V_{R1} = 20/3$ V

$V_{R2} = -10/3$ V ⟹ $i_{D1} = \frac{V_{R2}}{2k} < 0$ ✗

4) Consider the following circuit consisting of two ideal diodes

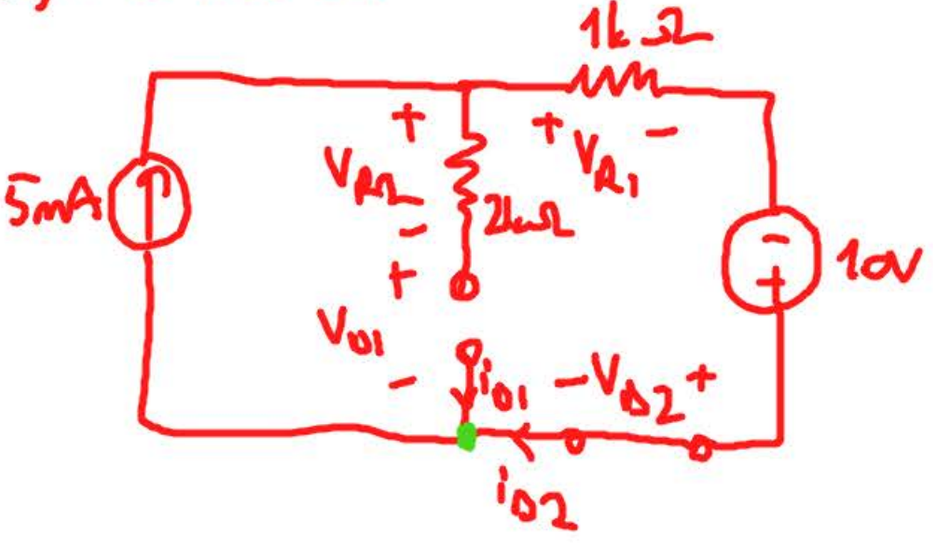
Find i_{D1} and V_{R2}.



Solution:

D1	D2
OFF	OFF
OFF	ON
ON	OFF
ON	ON

D) Assume D1 is OFF and D2 is ON



KCL: $i_{D1}+i_{D2}=5mA$

$i_{D1}=0 \Rightarrow i_{D2}=5mA$

Check whether $V_{D1}<0$ or not:

$V_{D1} = -V_{R2}+V_{R1}-10$

$= 0 + i_{D2}\cdot 1k - 10$

$= -5V < 0$ ✓